MAT316: Operations Research

Chapter : Simplex Method

Simplex Table

Equivalence of Maximization & Minimization Problem

Different Formations of Simplex Table

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Simplex Table

Minimize,

$$\begin{cases} x_1 \qquad\qquad +2x_4 \quad -x_5 = 10 \\ \qquad x_2 \qquad\quad -x_4 \quad -5x_5 = 20 \\ \qquad\qquad x_3 +6x_4 -12x_5 = 18 \\ \qquad\qquad\qquad -2x_4 \quad +3x_5 = 60 + z \end{cases}$$

will be written as:

	x_1	x_2	x_3	x_4	x_5	
x_1	1	0	0	2	-1	10
x_2	0	1	0	-1	-5	20
x_3	0	0	1	6	-12	18
	0	0	0	-2	3	60

This table is called simplex table of the current canonical form.

Current Basic Feasible Solution:

x_1	x_2	x_3	x_4	x_5	z
10	20	18	0	0	-60

Simplex Table: Detached Coefficients Notation

Problem

Minimize,

$$\begin{cases} x_1 \qquad\qquad +2x_4 \quad -x_5 = 10 \\ \qquad x_2 \qquad\quad -x_4 \quad -5x_5 = 20 \\ \qquad\qquad x_3 +6x_4 -12x_5 = 18 \\ \qquad\qquad\qquad -2x_4 \quad +3x_5 = 60 + z \end{cases}$$

Solve this problem using simplex tabular notation.

Solution:

Iteration-1:

	x_1	x_2	x_3	x_4	x_5	
x_1	1	0	0	2	-1	10
x_2	0	1	0	-1	-5	20
x_3	0	0	1	6	-12	18
	0	0	0	-2	3	60

Iteration-2:

	x_1	x_2	x_3	x_4	x_5	
x_1	1	0	$-\frac{1}{3}$	0	3	4
x_2	0	1	$\frac{1}{6}$	0	-7	23
x_4	0	0	$\frac{1}{6}$	1	-2	3
	0	0	$\frac{1}{3}$	0	-1	66

Iteration-3:

	x_1	x_2	x_3	x_4	x_5	
x_5	$\frac{1}{3}$	0	$-\frac{1}{9}$	0	1	$\frac{4}{3}$
x_2	$\frac{7}{3}$	1	$-\frac{11}{18}$	0	0	$\frac{97}{3}$
x_4	$\frac{2}{3}$	0	$-\frac{1}{18}$	1	0	$\frac{17}{3}$
	$\frac{1}{3}$	0	$\frac{2}{9}$	0	0	$\frac{202}{3}$

All terms in last row have non-negative coefficient: STOP.

Conclusion:

$$z_{\min} = -\frac{202}{3} \qquad \text{attained at} \qquad \left(0, \frac{97}{3}, 0, \frac{17}{3}, \frac{4}{3}\right).$$

Remark

For illustration purpose, we started with a given canonical form. if the problem was not in canonical form, we would have to convert the problem into a canonical form first so that we can start with a initial simplex table.

Problem

Maximize $z = 2x_1 + 3x_2 + 3x_3$ **subject to** $x_1, x_2, x_3 \geq 0$ **and**

$$\begin{cases} 3x_1 + 2x_2 \leq 60 \\ -x_1 + x_2 + 4x_3 \leq 10 \\ 2x_1 - 2x_2 + 5x_3 \leq 50 \end{cases}$$

Equivalent problem:

Minimize

$$z = -2x_1 - 3x_2 - 3x_3$$

subject to $x_1, x_2, x_3 \geq 0$ **and**

$$\begin{cases} 3x_1 + 2x_2 \leq 60 \\ -x_1 + x_2 + 4x_3 \leq 10 \\ 2x_1 - 2x_2 + 5x_3 \leq 50 \end{cases}$$

Introduce slack variables x_4, x_5, x_6. Write new constraints in standard form.

Minimize

$$-2x_1 - 3x_2 - 3x_3 = z$$

subject to $x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$ **and**

$$\begin{cases} 3x_1 + 2x_2 + x_4 = 60 \\ -x_1 + x_2 + 4x_3 + x_5 = 10 \\ 2x_1 - 2x_2 + 5x_3 + x_6 = 50 \end{cases}$$

Canonical form! Note the convenience of adding slack variables.

Note

- The constraints are already in canonical form.
- The basic solution is feasible.
- The bojective function is already in terms of the non-basic variables.

Iteration-1:

	x_1	x_2	x_3	x_4	x_5	x_6	
x_4	3	2	0	1	0	0	60
x_5	-1	1	4	0	1	0	10
x_6	2	-2	5	0	0	1	50
	-2	-3	-3	0	0	0	0

Iteration-2:

	x_1	x_2	x_3	x_4	x_5	x_6	
x_4	5	0	-8	1	-2	0	40
x_2	-1	1	4	0	1	0	10
x_6	0	0	13	0	2	1	70
	-5	0	9	0	3	0	30

Iteration-3:

	x_1	x_2	x_3	x_4	x_5	x_6	
x_1	1	0	$-\frac{8}{5}$	$\frac{1}{5}$	$-\frac{2}{5}$	0	8
x_2	0	1	$\frac{12}{5}$	$\frac{1}{5}$	$\frac{3}{5}$	0	18
x_6	0	0	13	0	2	1	70
	0	0	1	1	1	0	70

Optimal solution of the minimization problem:

x_1	x_2	x_3	x_4	x_5	x_6	z_{min}
8	18	0	0	0	70	-70

Optimal solution of the maximization problem:

x_1	x_2	x_3	x_4	x_5	x_6	z_{max}
8	18	0	0	0	70	70

Back to the original problem:

At $x_1 = 8$, $x_2 = 18$, $x_3 = 0$, we get the max value of 70.

Remark

- Adding slack variables could put LP in canonical form.
- Since $x_4 = 0, x_5 = 0$, the first two constraints have no slack.
- Since $x_6 > 0$, the 3rd constraint is satisfied with strict inequality $<$.
- LP must be in canonical form to set up the initial simplex table.
- Algorithm ends when the optimal solution is found or the problem is unbounded.

Formation of Simplex Table in Different Mannersa

An Alternate Way:

For example, the form

$$\begin{aligned} z - 2x_1 - 3x_2 \qquad\qquad &= 0 \\ 2x_1 + \ \ x_2 + s_1 \qquad &= 4 \\ x_1 + 2x_2 \qquad\quad + s_2 &= 5 \end{aligned}$$

would be written in abbreviated form as shown in the following table.

Basic	x_1	x_2	s_1	s_2	RHS
z	-2	-3	0	0	0
s_1	2	1	1	0	4
s_2	1	2	0	1	5

Another Way:

	$C_j \rightarrow$	5	2	3	-1	1		
Basis	C_B	x_1	x_2	x_3	x_4	x_5	*constants*	*ratio*
x_3	3	$1/2$	1	1	$1/2$	0	4	8
x_5	1	$5/2$	3	0	$-1/2$	1	3	$6/5$
	CR row	1	-4	0	-2	0	$Z = 15$	